ANUSHEEL CHAPAGAIN, MASc, BEng

ENGINEERING | DATA | PRODUCT DEVELOPMENT

St. John's, NL, Canada | Authorized to work in Canada
+1 709 691 3985 | chapagainanusheel@gmail.com
linkedin.com/in/chapagainanusheel | github.com/anusheeel | anusheeel.github.io

PROFILE

Engineer and hands-on builder with about five years of experience across energy systems, construction project controls, data engineering, mechanical design and early-stage software. My work usually starts with a practical problem: I talk with the people doing the work, understand where the process is breaking down, and build a tool they can actually use. Recent work includes construction reporting systems, energy forecasting, LiDAR/CAD workflows, and internal tools for cost, equipment, purchasing and workforce data. I want to keep combining technical problem-solving with product decisions, working closely with stakeholders and hands-on delivery.

CORE STRENGTHS

Practical product building: talking with users, mapping workflows, defining requirements and acceptance criteria, building prototypes, running demonstrations, onboarding people and improving the product from their feedback.

Project controls and operations: cost history, progress, labour and productivity, equipment, purchasing, field reporting, inspection, commissioning, schedules, risks and KPI reporting.

Data and software: Python, SQL, JavaScript, Power BI, Power Query, Excel, pandas, ETL, data modelling, validation, REST APIs and internal web tools.

AI, cloud and analytics: XGBoost, scikit-learn, forecasting, time-series analysis, human review of AI-assisted workflows, AWS, PostgreSQL/Supabase, Snowflake, Quickbase and Oracle.

Engineering and energy: building-energy audit workflows, utility-bill and weather analysis, LiDAR/GLB workflows, CAD, Fusion 360, FEA, prototyping and distributed-energy systems.

PROFESSIONAL EXPERIENCE

GREEN INFRASTRUCTURE PARTNERS | Project Controls Coordinator

St. John's, NL | Jun 2025-Present

- Work with the VP of Cost and Project Controls and teams across project controls, equipment, purchasing, HR, finance and field operations to understand recurring problems and turn them into practical reporting tools and workflows.
- Automated reporting for the Eastern Kings Wind Power project using Python, SQL and Power BI, replacing siloed spreadsheets with current progress, cost and risk information.
- Built a searchable historical cost-estimate and budget application, making it easier for teams to find and reuse past project information.
- Developed browser-based equipment and purchasing workflows covering project assignment, daily hours and kilometres, maintenance and registration dates, approvals, vendors, status and lifecycle history.
- Built a Python ETL process to consolidate timesheets for 70+ employees with data-quality checks, and continue moving legacy Excel reporting into structured Power BI models.

ENERGY INTELLIGENCE | Founding Engineer

St. John's / Remote, Canada | Nov 2024-Present

- Lead the architecture and product development of a building-energy audit platform that connects building capture, utility bills, forecasting, advisor review and customer recommendations.
- Work directly with energy advisors to understand how audits happen in practice, then use their feedback to improve the product; the platform has been used in onboarding and testing with 20+ advisors.
- Built and migrated a Python, AWS and XGBoost energy-bill forecasting pipeline that reached approximately 80% accuracy in project testing.
- Designed and developed a browser-based CAD/GLB tool for LiDAR-scanned buildings, including measurement, drawing, extrusion, roof modelling, annotation and volume calculation.
- Added manual correction workflows for rooms, appliances and complex roof structures that scans can miss, so advisors can finish a reliable building model without leaving the platform.
- Built portfolio and project workflows that connect multiple homes, mobile field scanning, work status and energy modelling; secured $10,000 in early-stage funding through product demonstrations.

ATKINSRÉALIS (SDP GP) | Analytics / Data Engineer

Argentia, NL | Aug 2023-Jan 2025

- Developed a full-stack reporting tool for the multi-billion-dollar West White Rose construction project, improving progress-tracking efficiency by 40% for area managers and engineers.
- Automated processing for 30,000+ labour-hour records and reduced recurring report preparation by 75% using Python, Snowflake, Excel and Power BI.
- Modelled a digital-twin database for 200,000+ construction documents and thousands of revisions, with role-based access for 70+ non-technical users across several organizations.
- Built APIs between legacy Oracle systems and modern web tools, reducing manual data transfer time by 50% and improving the consistency of recurring reporting.
- Developed machine-learning models for labour allocation and fatigue or safety-risk patterns across 1,000+ workers using pandas and scikit-learn.
- Worked across project controls, construction, finance, scheduling, quality, commissioning and leadership; supported adoption through demonstrations, documentation, training and day-to-day user support.

RE/MAX INFINITY | System Analyst

Greater St. John's Area | Jan 2023-Aug 2023

- Helped real-estate agents use and troubleshoot kvCORE CRM, portfolio websites and backend advertising data.
- Supported campaign setup, lead collection and SEO work, translating technical issues into clear steps for non-technical users.

ECOBLOC (TECHSTARS '22) | Mechanical Design Engineer

St. John's, NL | Aug 2022-Feb 2023

- Designed the enclosure and mechanical assembly for rodent-repelling technology, producing Fusion 360 models and engineering drawings and iterating with a cross-functional startup team.
- Used FEA, tolerance studies, prototype testing and performance data to improve structural integrity, manufacturability, assembly and material cost.

GAJJABIT | Energy Data Engineer

Kathmandu, Nepal | Dec 2019-Nov 2021

- Researched 500,000 household energy-consumption records and engineered time-series features to understand usage and seasonal demand patterns.
- Built Python, SQL and pandas workflows for ETL, model training and visualization, and used scikit-learn to forecast household demand and improve load-shedding schedules.

RESEARCH AND SELECTED PROJECTS

Peer-to-peer energy trading: Co-authored *"Developing an Appropriate Energy Trading Algorithm and Techno-Economic Analysis between Peer-to-Peer within a Partly Independent Microgrid,"* published in Energies, 2023, 16(3), 1549. The work covered a supply-demand-ratio pricing approach and dynamic matchmaking logic for local energy trading.

Building-energy research: Co-authored an undergraduate thesis, "A Study on Use of Attic Space for Comfort Air Conditioning" (2019), examining how trapped roof-space heat could be used in residential thermal systems.

Published iOS products: Designed, built and published three iPhone apps: GigReady for setlists and practice planning, Multi Sound Metronome for rhythm practice, and TinyHours for simple, local-first baby-care tracking. I took each product from a practical user problem through design, development and App Store release.

Independent software builds: Ongoing energy and AI experiments include AI Radio, an energy-demand predictor, GridWatch and a weather forecaster. Selected work is available on GitHub.

EDUCATION AND PROFESSIONAL DEVELOPMENT

Memorial University of Newfoundland **2023**
MASc, Energy Systems Engineering

Kathmandu University **2019**
BEng, Mechanical Engineering

techNL **2026**
AI for Data Engineering